

Industrial Internship Report on **Crop Yield Prediction Using Machine Learning**

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Executive Summary

This report presents the work completed during my six-week Industrial Internship in the domain of Data Science and Machine Learning, organized by Upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT).

The internship focused on understanding the complete Machine Learning workflow through practical implementation. As part of the internship, I developed a Crop Yield Prediction Using Machine Learning project using Python and Google Colab. The project involved data preprocessing, exploratory data analysis, feature engineering, model training using the Random Forest Regressor algorithm, and performance evaluation.

The developed model achieved an R^2 Score of 0.9066 with a Mean Absolute Error (MAE) of 10.43, indicating good prediction accuracy. This internship significantly enhanced my technical knowledge, analytical thinking, and practical understanding of Machine Learning.

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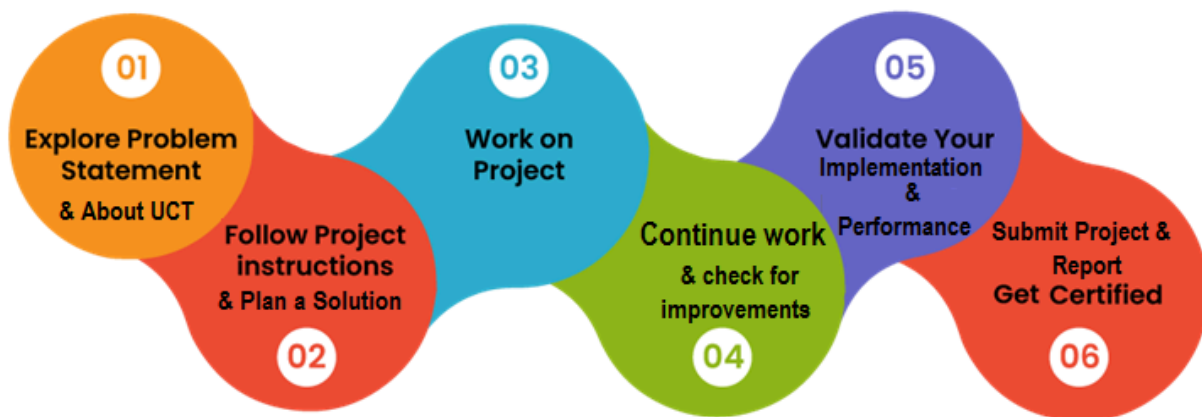
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1 Preface

Industrial internships play a significant role in bridging the gap between academic learning and real-world industry practices. This six-week internship provided me with an opportunity to understand practical applications of Data Science and Machine Learning while working on a real-world problem.

Throughout the internship, I learned Python programming, NumPy, Pandas, Probability and Statistics, Machine Learning concepts, data preprocessing, model building, and performance evaluation. These concepts helped me successfully develop a Crop Yield Prediction model.

How Program was planned



I sincerely thank UniConverge Technologies Pvt. Ltd., The IoT Academy, Upskill Campus, my college faculty, and all the mentors for providing guidance and support throughout the internship. This internship has greatly improved my technical skills and increased my confidence in solving real-world problems using Machine Learning.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.**



i. UCT IoT Platform(uct Insight)

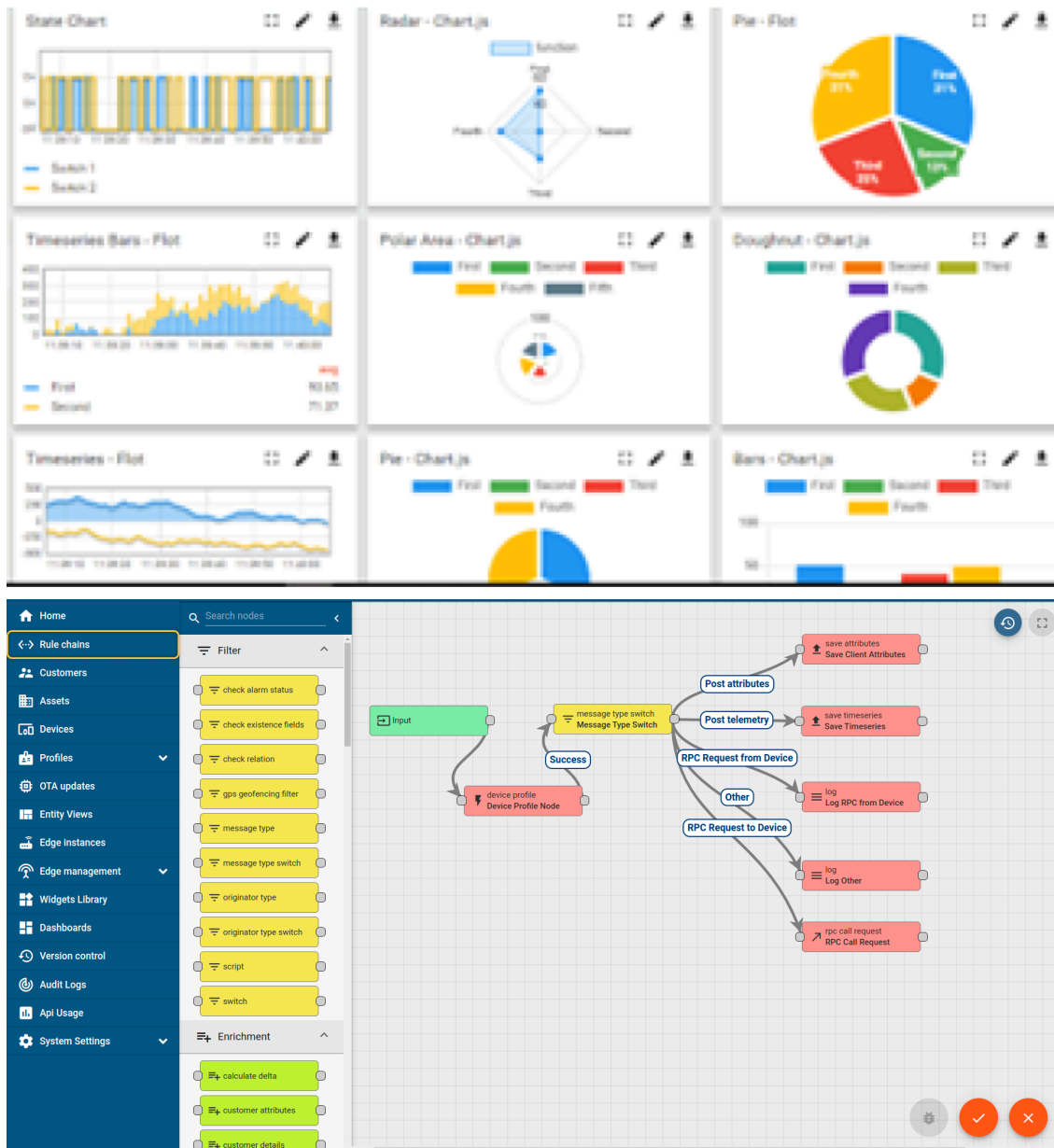
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



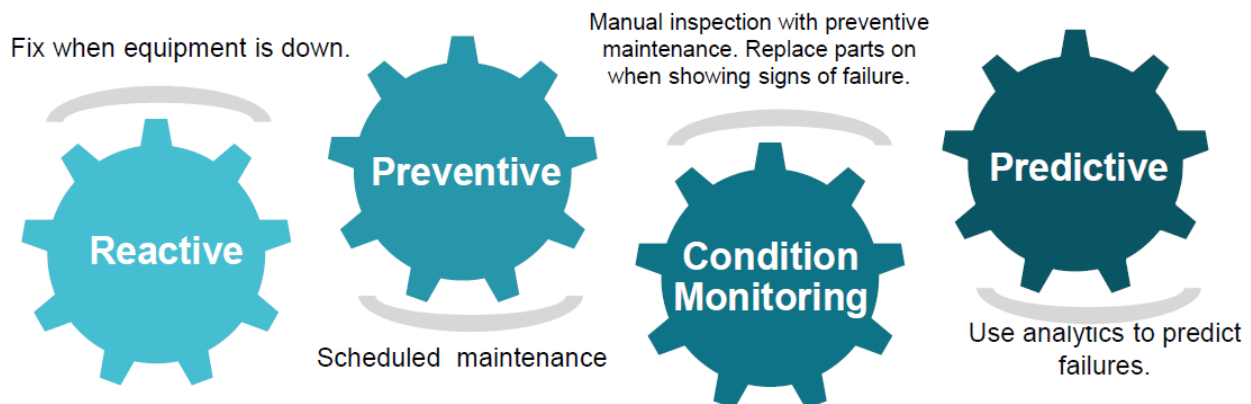


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

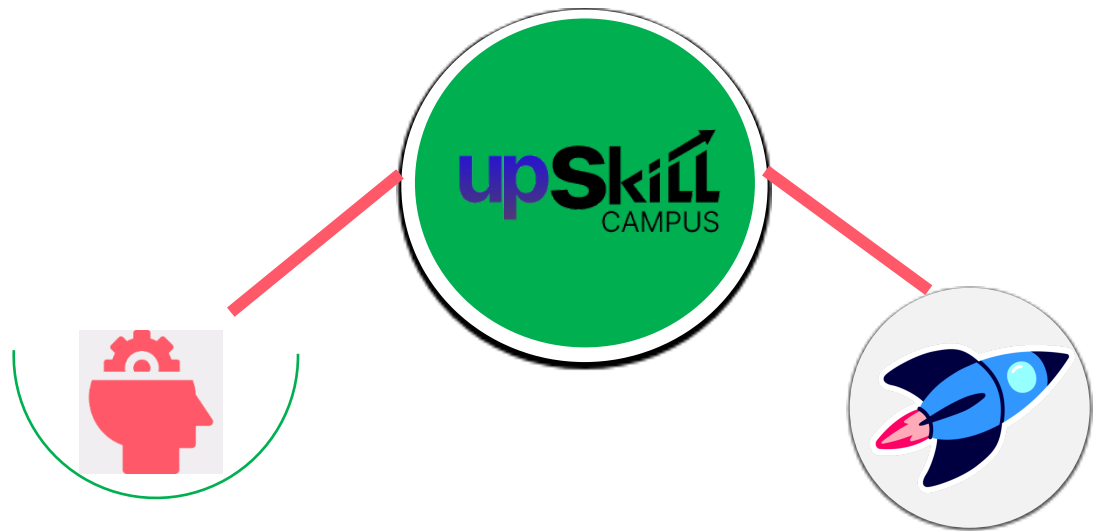
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>

Career growth/upskilling

- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking

- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform

- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform

- Job portal
- Internship portal
- Freelancing projects

nsi

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

Agriculture plays an important role in the economy, and predicting crop yield accurately can help farmers make informed decisions. Traditional prediction methods often fail to consider multiple influencing factors simultaneously.

The objective of this project is to develop a Machine Learning model that predicts crop yield using historical agricultural data. The model considers various features such as crop type, season, state, cultivated area, annual rainfall, fertilizer usage, pesticide usage, and production to estimate crop yield with high accuracy.

4 Existing and Proposed solution

Existing Solution

Traditional crop yield prediction methods rely mainly on manual calculations, historical averages, and expert knowledge. These methods often produce less accurate results due to changing environmental conditions.

Limitations

- Low prediction accuracy
- Time-consuming manual analysis
- Unable to process large datasets efficiently
- Limited scalability

Proposed Solution

The proposed solution uses Machine Learning to predict crop yield automatically. Historical agricultural data is preprocessed and used to train a Random Forest Regression model. The trained model predicts crop yield with high accuracy and can assist in agricultural decision-making.

4.1 Code submission (Github link):
<https://github.com/anweshajena23-stack/Upskill-campus/tree/main>

4.2 Report submission (Github link) :
https://github.com/anweshajena23-stack/Upskill-campus/blob/main/Crop_%20prediction%20USC_UCT.pdf

5 Proposed Design/ Model

The project follows the Machine Learning workflow:

- Data Collection
- Data Preprocessing
- Missing Value Handling
- Label Encoding of Categorical Variables
- Feature Selection
- Train-Test Split
- Random Forest Regression Model Training
- Model Prediction
- Performance Evaluation using R^2 Score and MAE

Algorithm Used: Random Forest Regressor

Development Platform: Google Colab

Programming Language: Python

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

The model was trained using 80% of the dataset and tested using the remaining 20%.

6.2 Test Procedure

- Loaded the dataset.
- Performed preprocessing.
- Encoded categorical variables.
- Trained the Random Forest model.
- Predicted crop yield.
- Evaluated model performance.

6.3 Performance Outcome

R² Score: 0.9066

Mean Absolute Error: 10.43

The model achieved high prediction accuracy and demonstrated its effectiveness in estimating crop yield using historical agricultural data.

7 My learnings

During this internship, I gained practical knowledge in Data Science and Machine Learning. I learned Python programming, conditional statements, NumPy, Pandas, Probability and Statistics, Machine Learning concepts, data preprocessing, feature engineering, Random Forest Regression, model evaluation techniques, and data visualization.

I also became familiar with Google Colab, GitHub, and project documentation. This internship strengthened my analytical thinking, programming skills, and problem-solving abilities, preparing me for future opportunities in Data Science and Machine Learning.

8 Future work scope

The project can be further improved by:

- Using larger and more diverse datasets.
- Integrating real-time weather information.
- Comparing multiple Machine Learning algorithms.
- Developing a web application using Flask or Streamlit.
- Deploying the model on cloud platforms for real-time prediction.
- Enhancing prediction accuracy through advanced feature engineering and hyperparameter tuning..